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1 /Users/mehakagrawal/PycharmProjects/
  Final_Dissertation/.venv/bin/python /Users/
  mehakagrawal/Desktop/Final_Dissertation/Codes/8.
  q2_age_tf_activity_and_comparison/
  q2_age_tf_activity_and_comparison.py
2 Full TF-activity comparison: mutant (Q1) versus
  natural ageing (Q2)
3
4 Downloading CollecTRI regulons (human) via
  decoupler/OmniPath...
5   CollecTRI: 1185 TFs, 42990 interactions.
6
7 Scoring natural ageing (Q2) across all CollecTRI
  TFs...
8   Human gene-level vector (ageing): 13167 symbols.
9   Saved: tf_activity_age.csv
10  TFs scored: 698; significant (p<0.05): 37
11  Most activated in ageing: HNF1A, HNF4A, CEBPG,
  NR1I3, NR1H3, CDX2, NR1H4, CEBPA
12  Most repressed in ageing: NR1D1, ZNF202, TEAD2,
  EHF, HESX1, NFATC1, GRHL2, NFATC4
13
14 Q1 mutant TF activities: 698 TFs, 71 significant.
15
16 Comparison over 698 TFs scored in both contrasts:
17   Pearson r = -0.145 (p = 1.2e-04)
18   Same direction: 301 of 698 TFs (43%)
19   Significant in BOTH: 5 TFs
20     of those, same direction: 40%
21
22 Key transcription factors (positive = activated,
  negative = repressed):
23   E2F1    mutant -7.604 (p=0.000)    ageing +0.320 (p
    =0.926)    (different direction)
24   E2F3    mutant -4.058 (p=0.001)    ageing -0.847 (p
    =0.767)    (same direction)
25   E2F4    mutant -6.014 (p=0.000)    ageing +0.242 (p
    =0.946)    (different direction)
26   TP53    mutant +6.417 (p=0.000)    ageing -0.782 (p
    =0.790)    (different direction)
27   AP1     mutant +6.536 (p=0.000)    ageing +3.212 (p

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27 =0.029)    (same direction)
28  JUN      mutant +5.488 (p=0.000)    ageing +0.150 (p
    =0.971)    (same direction)
29  NFkB     mutant +5.007 (p=0.000)    ageing -0.008 (p
    =0.999)    (different direction)
30  HIF1A    mutant +5.722 (p=0.000)    ageing +0.164 (p
    =0.971)    (same direction)
31  SMAD3    mutant +4.924 (p=0.000)    ageing -0.792 (p
    =0.790)    (different direction)
32  MYC      mutant +1.651 (p=0.344)    ageing +0.409 (p
    =0.906)    (same direction)
33
34 Saved: q1_vs_age_tf_comparison.csv
35 Saved: q1_vs_age_scatter.png
36
37 Outputs in: /Users/mehakagrawal/Desktop/
    Final_Dissertation/Outputs/8.
    q2_age_tf_activity_and_comparison
38
39 Process finished with exit code 0
40
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